

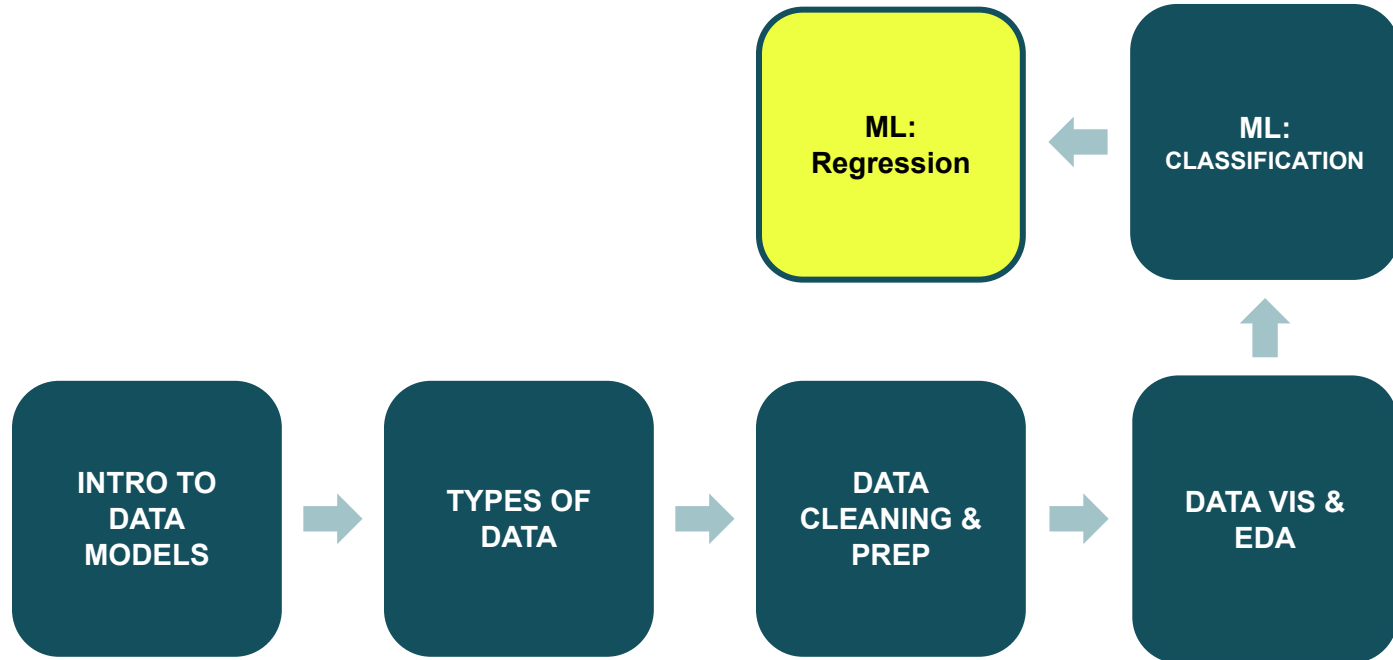
ML: Regression

CPE 232: Data Models

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Review



Outline

- Recap
- Intro to Regression
- Types of Regression
- Evaluation

Decision Tree

Classification

Node Impurity

Entropy

Information Gain

Training

Testing

Evaluation

Precision

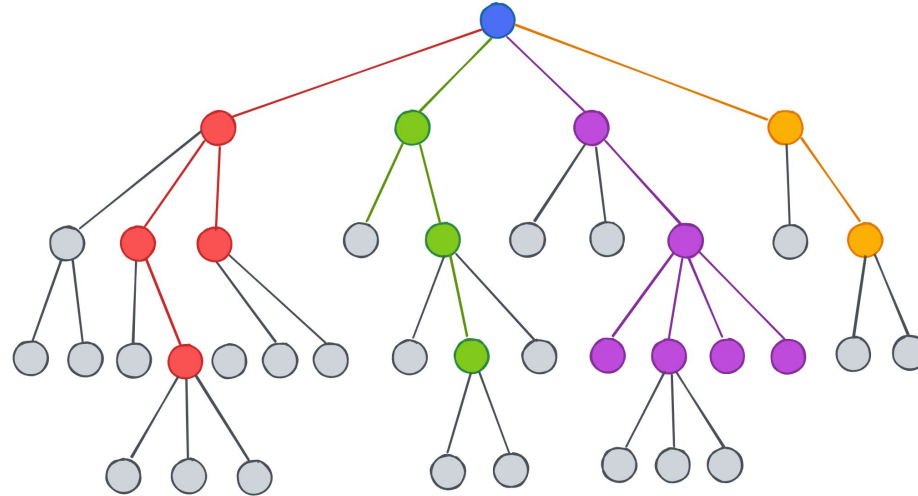
Recall

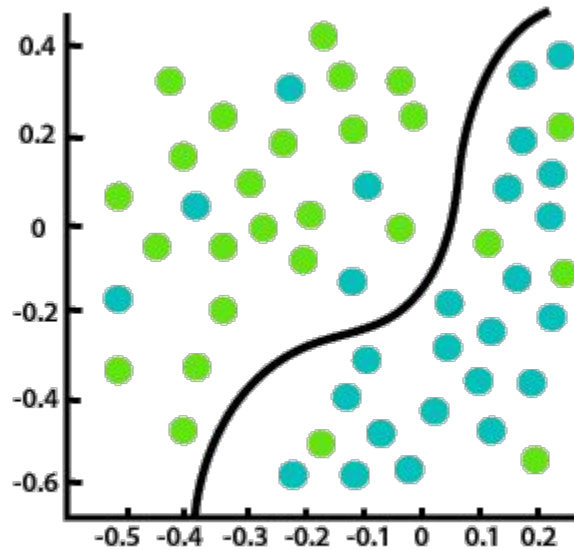
F-measure

Confusion Matrix

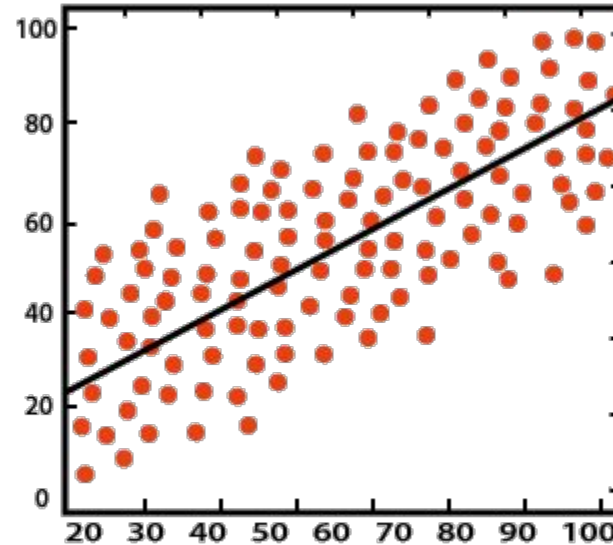
Learnable

Hyperparameter





Classification



Regression

Ref: <https://www.javatpoint.com/regression-vs-classification-in-machine-learning>

Real world input

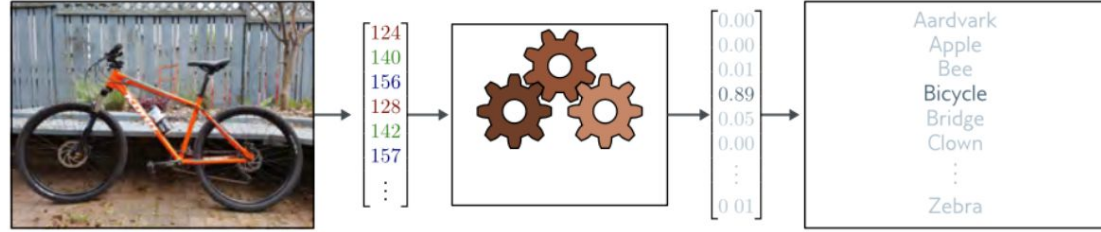
Model input

Model

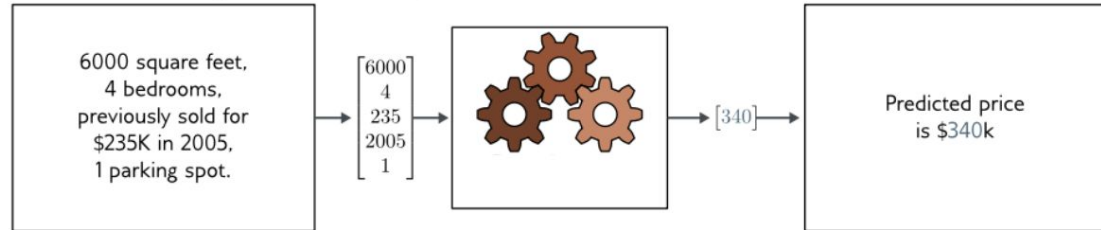
Model output

Real world output

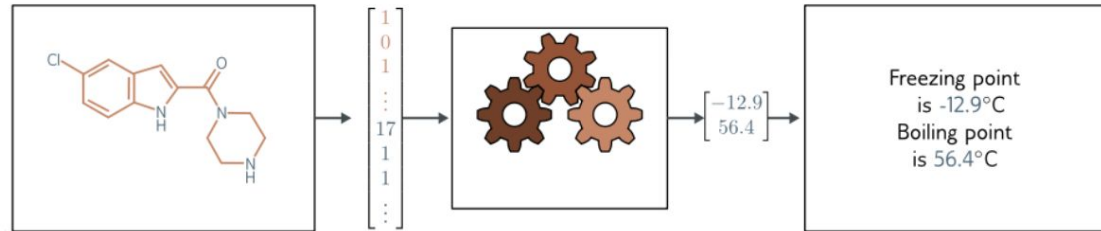
Image recognition (Classification)



House price prediction (Regression)



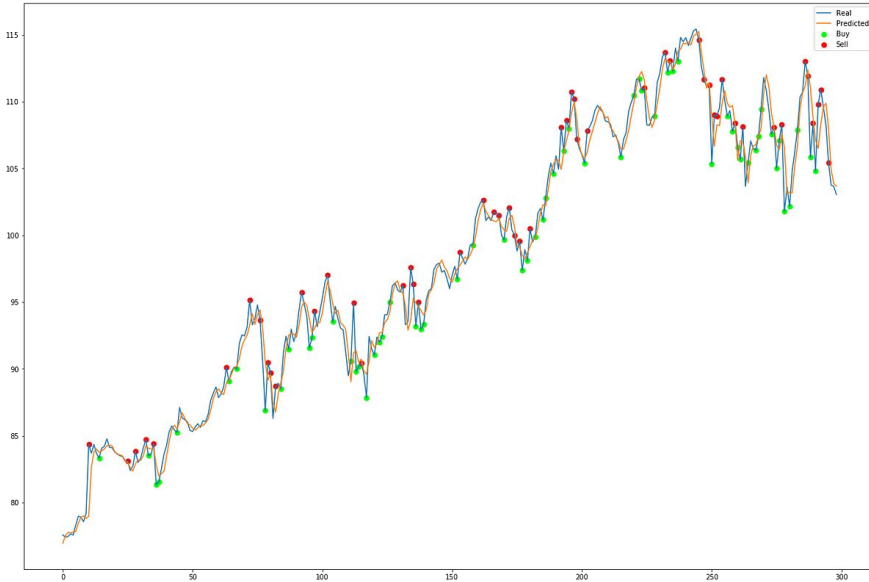
Chemical structure characterization (Regression)



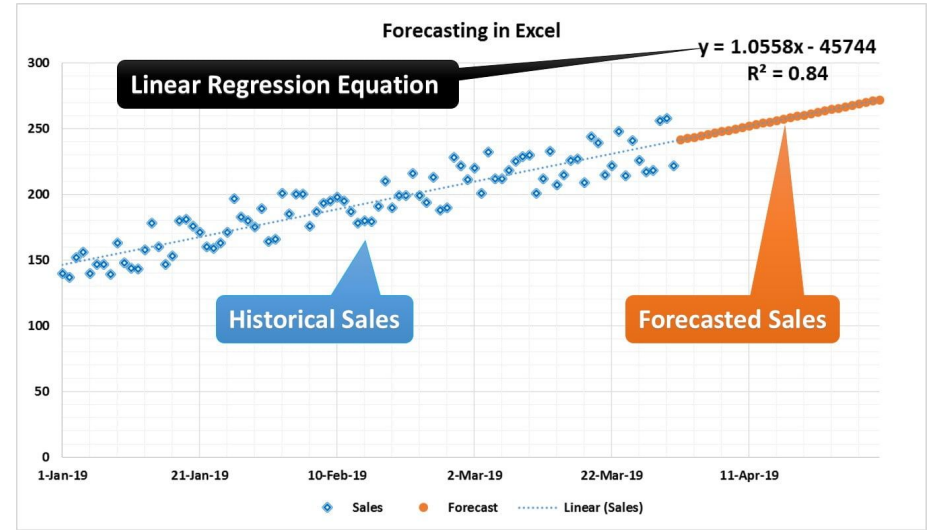
Regression

- A predictive modeling technique
- To estimate the relationship between *two* kinds of variables
 - Independent variable(s) $x_1, x_2, \dots, x_n$ → Predictors
 - Dependent variable y → Outcome
- Has an infinite number of possible outcomes
- **Goal:** To predict the value of the dependent variable based on the values of the independent variables

Regression: Examples



Stock Price Prediction



Sales Forecasting

Time-series

- Healthcare
- Academic
- Environmental Science
- Manufacturing
- etc.

Regression: Importance

- Understand relationships among data points
- Prediction (Strategy, Planning, etc.)
- Decision making
- Optimization

Data

Attributes

Records

Attribute 1	Attribute 2	...	Attribute k	Target
x_{11}	x_{12}	...	x_{1k}	y_1
x_{21}	x_{22}	...	x_{2k}	y_2
x_{31}	x_{32}	...	x_{3k}	y_3
...	...	...	...	...
x_{n1}	x_{n2}	...	x_{nk}	y_n

Values to predict

Types of Regression

- Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression
- Decision Tree Regression
- And more...

General Linear Model

$$y = mx + c$$

- ❑ Predict the Final Exam scores (y)
- ❑ How many hours put into studying (x)
- ❑ Baseline:
scores = 0.5(#hours) + 10

Data = Model + Error

Linear Regression

- A statistical method used to model the relationship between:

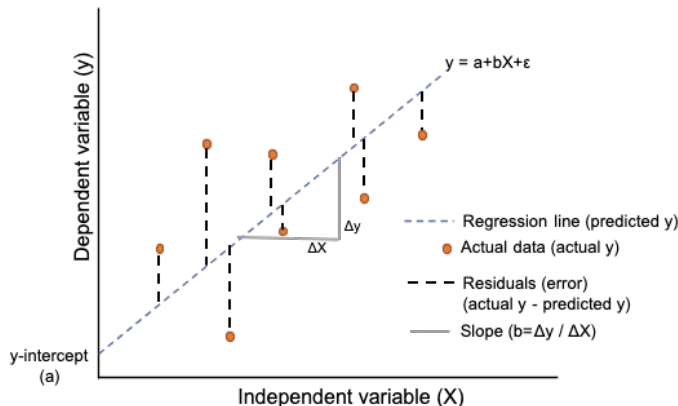
- A dependent variable
- One or more independent variables

$$y$$
$$x_1, x_2, \dots, x_n$$

↪ Multiple Linear Regression

- A sloped straight line describing the relationship within the variables $y = mx + c$

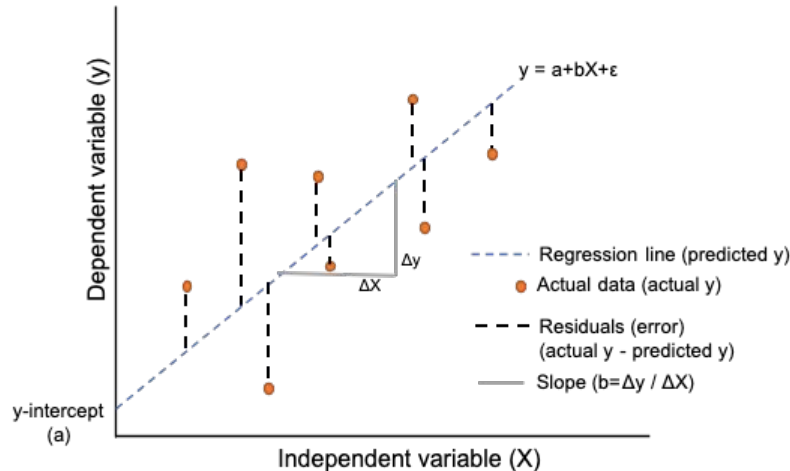
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_n x_n + \dots + \epsilon$$



y	Dependent variable to predict
$x_1, x_2, \dots, x_n$	Independent variables (predictors or features)
β_0	Intercept
$\beta_1, \beta_2, \dots, \beta_n$	Coefficients (aka slopes)
ϵ	Error term

Linear Regression (Cont.)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_n x_n + \dots + \epsilon$$

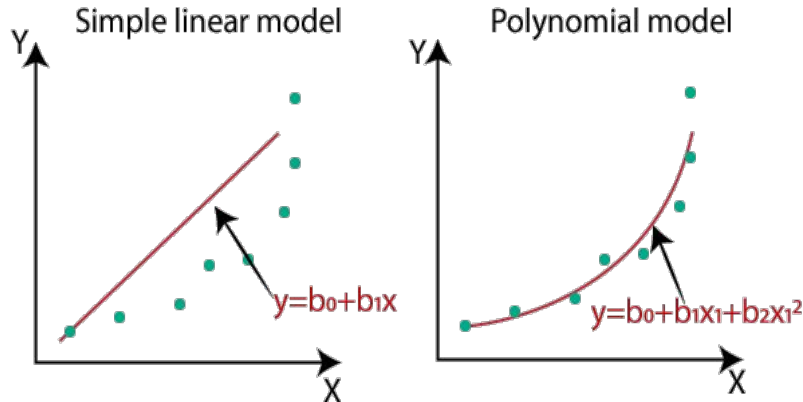


Goal:

- Estimate values of coefficients that **best fit** the observed data $\beta_1, \beta_2, \dots, \beta_n$
- **Minimize the differences** between the observed and predicted values of the dependent variable y
- **Sum of squared differences**
 - aka **SSE**: Sum of Squared Error
 - To be minimized

Polynomial Regression

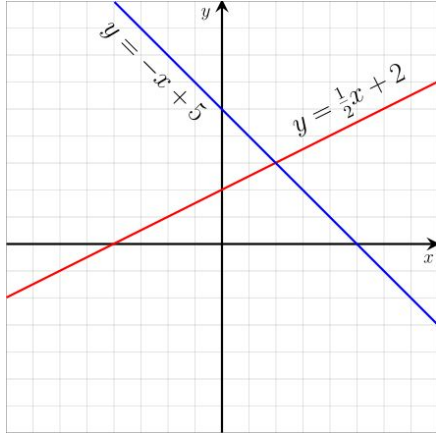
- Modeling more complex relationships between
 - A dependent variable y
 - One or more independent variables $x_1, x_2, \dots, x_n$
- n-th degree polynomial function $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_n x^n + \epsilon$



- Useful when the relationship between variables is not linear (aka nonlinear)
- Prone to overfitting
- More difficult to interpret
- Selecting the degree of the polynomial is therefore very important

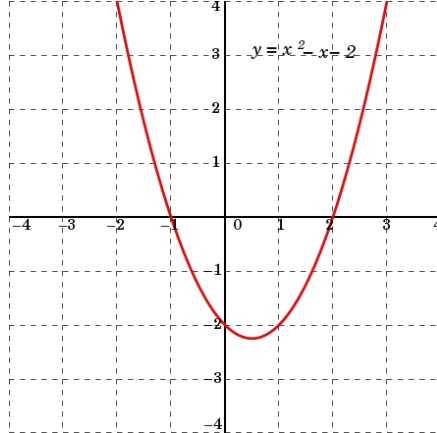
Different Degrees

Degree 1



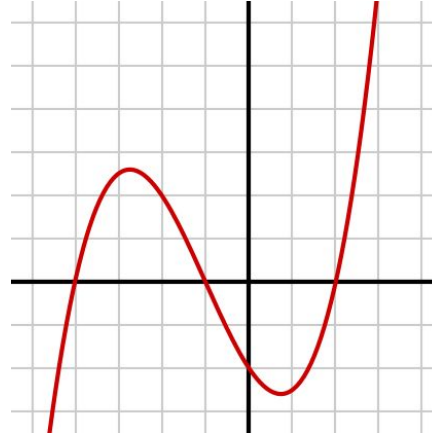
Linear

Degree 2



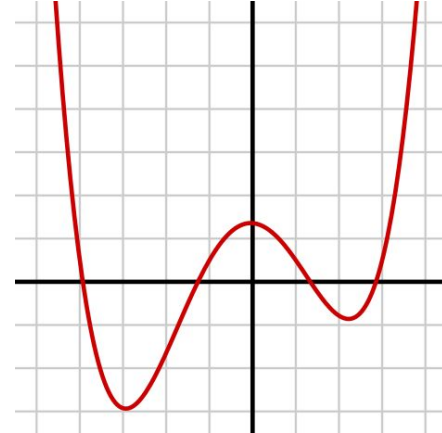
Quadratic

Degree 3



Cubic

Degree 4

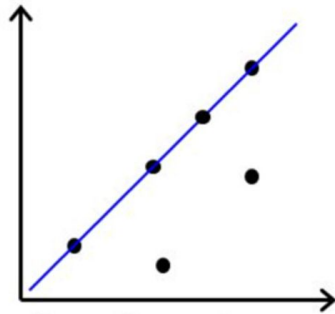


Quartic

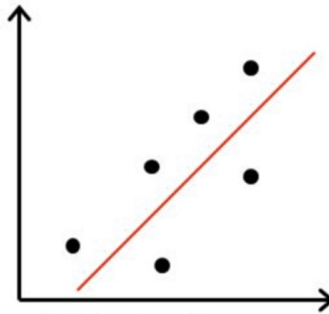
Best Line

Objective: Want to fit the "best" line to the data points (that exhibit linear relation).

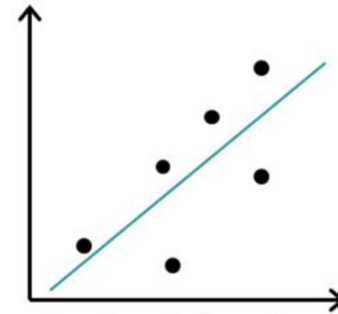
– How do we define "best"?



Pass through as many points as possible



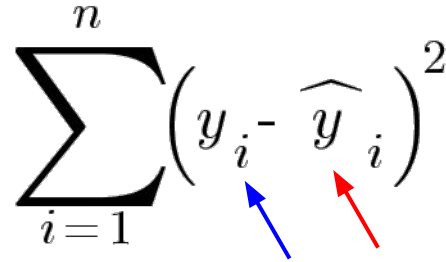
Minimize the maximum residual of each point



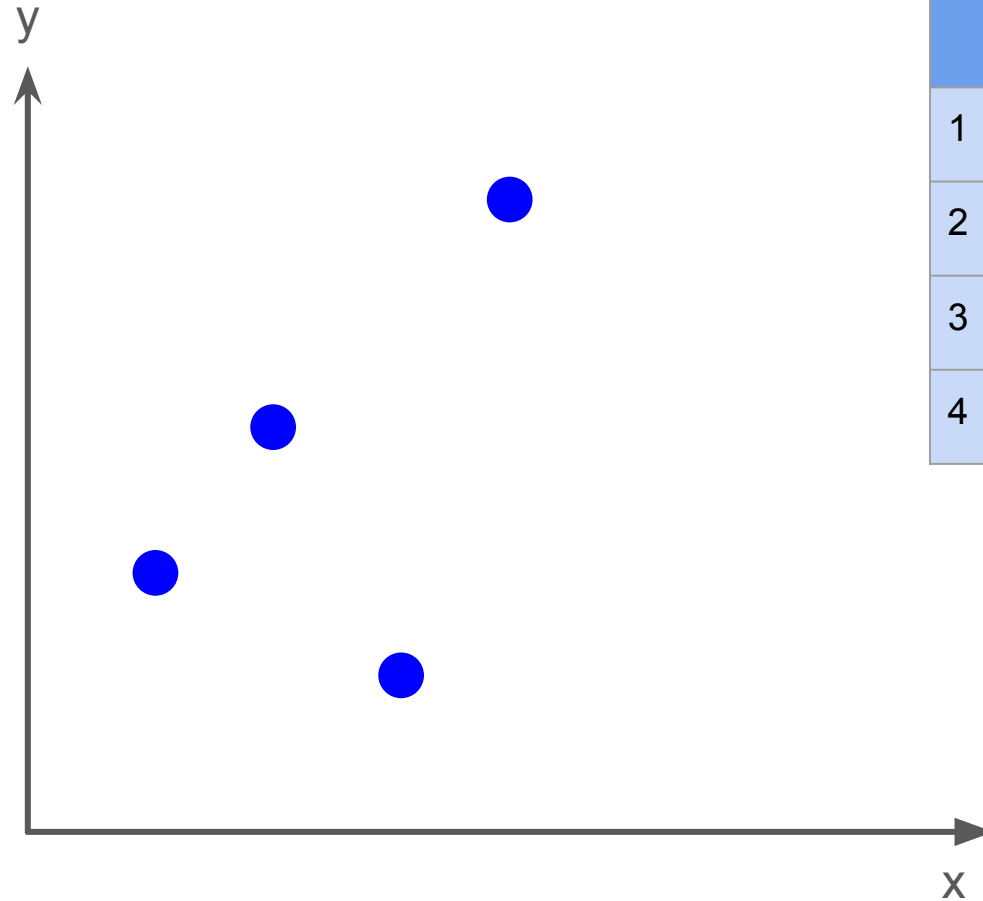
Each point carries the same weight

SSE: Sum of Squared Error

- Quantifies the difference between actual target values and predicted values
 - y_true** (actual value)
 - y_pred** (predicted value)

$$\sum_{i=1}^n \left(y_i - \widehat{y}_i \right)^2$$
The diagram shows the mathematical formula for the Sum of Squared Error (SSE). It consists of a summation symbol with a subscript 'i=1' and a superscript 'n'. Inside the summation is the expression (y_i - y-hat_i)^2. A blue arrow points from the text 'y_true' in the list above to the 'y_i' term in the formula. A red arrow points from the text 'y_pred' in the list above to the 'y-hat_i' term in the formula.

- The model's objective is to minimize error, and improve accuracy



Compute SSE

- Predict y from x for a regression problem
- Use Linear Regression Model

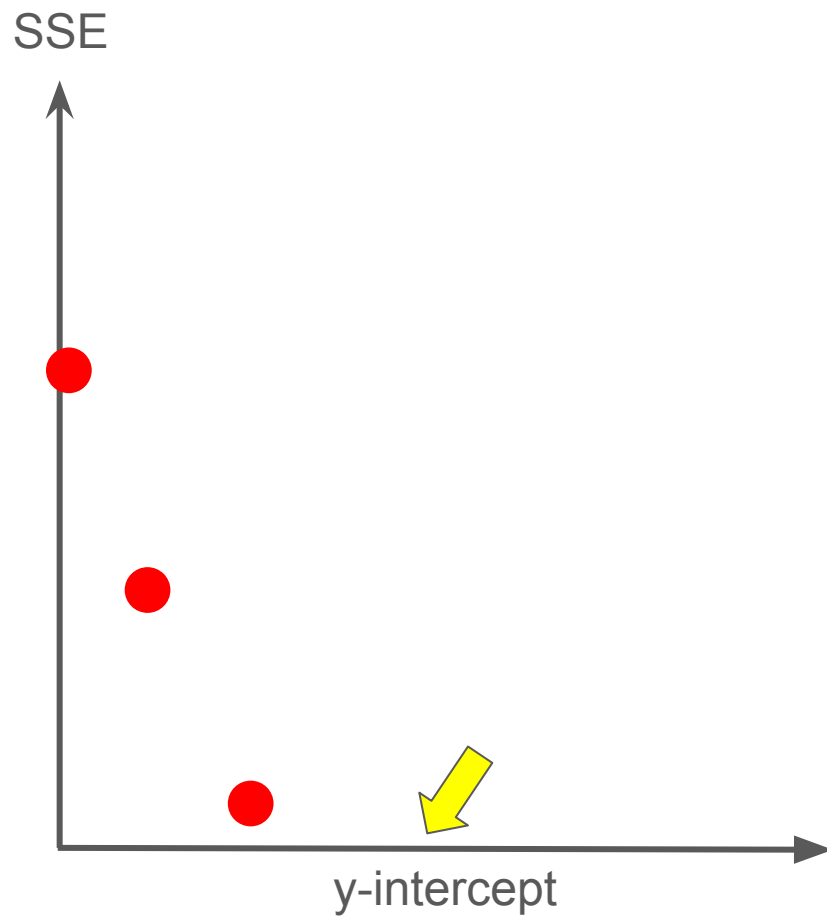
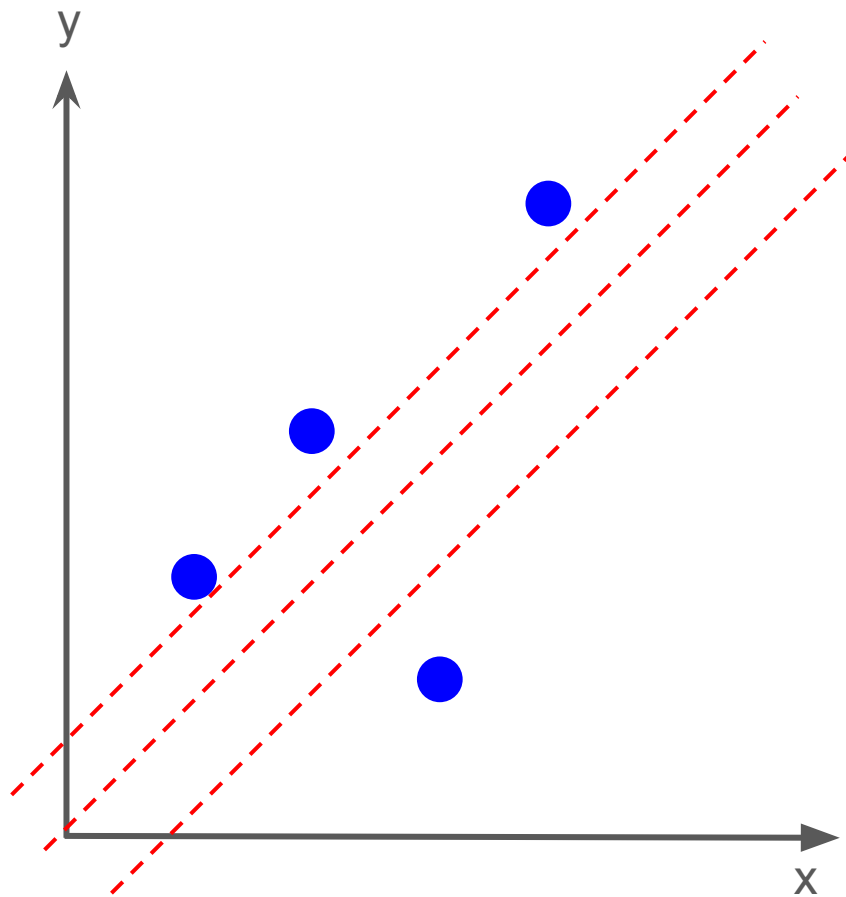
$$y_{pred} = wx + b$$

- **y_pred** → predicted value
- **w** (weight) and **b** (bias) are trainable parameters
- What should be **w** and **b** that minimize SSE?

y_true	y_pred	Error	Error ²
3.0	2.5	0.5	0.25
5.0	5.4	-0.4	0.16
2.0	2.2	-0.2	0.04
8.0	7.8	0.2	0.04

Computing SSE:

$$\begin{aligned} &= (0.25 + 0.16 + 0.04 + 0.04) \\ &= 0.49 \end{aligned}$$



Minimize Loss

- Minimize error via an optimization algorithm
 - Loss function: SSE, MSE, RMSE, MAE, and more..
 - Optimization algorithm: Gradient Descent
- Take a derivative of the loss function
 - Intercept (*i.e.* b)
 - Slope (*i.e.* w)
- Other parameter
 - Learning rate

Loss Function (Cost Function)

n = number of samples

y_i = actual (observed) value

$\hat{y}_i$ = predicted value

Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Gradient Descent



Gradient Descent

- **Gradient** → Slope
- **Descent** → Going down
- **Mechanism** → Following the *gradient* (slope) of the loss function to *descend* (go down) to the lowest point, or minimum error.

Analogy

1. Feel the slope (Find direction)
2. Take a step (Small step)
3. Repeat (Till reaching the bottom)

Gradient Descent Components

1. Loss Function (Landscape)
2. Learning Rate (Step size)
3. Convergence (Goal)

Minimize Loss - How?

Loss Function

- Which loss function is used?
- Minimize it
- Example: use MSE

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$J(w, b) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Regression Model

- Start with random parameters w, b
- Compute the error
- Move parameters in the direction that reduces the error
- Repeat until the error stops decreasing

$$y_{pred} = wx + b$$

Update Parameters

- Via derivatives
- Take a derivative of the loss function
 - Intercept (i.e. b)
 - Slope (i.e. w)
- Subtract the gradient
 - To *move downhill* toward the minimum
- Other parameter
 - $\alpha \rightarrow$ Learning rate (aka step size)
 - Too small $\rightarrow$ learning is very slow
 - Too large $\rightarrow$ may overshoot the minimum

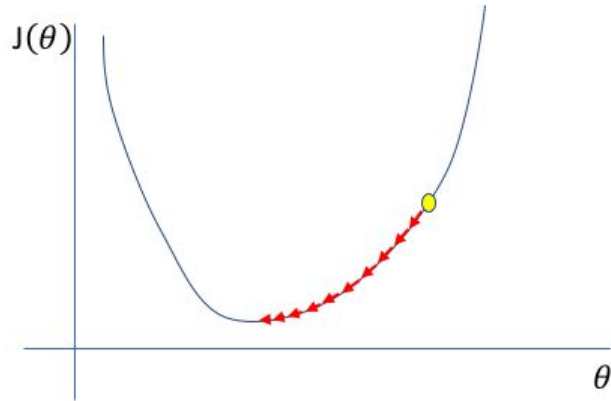
$$J(w, b) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$w = w - \alpha \frac{\partial J}{\partial w}$$

$$b = b - \alpha \frac{\partial J}{\partial b}$$

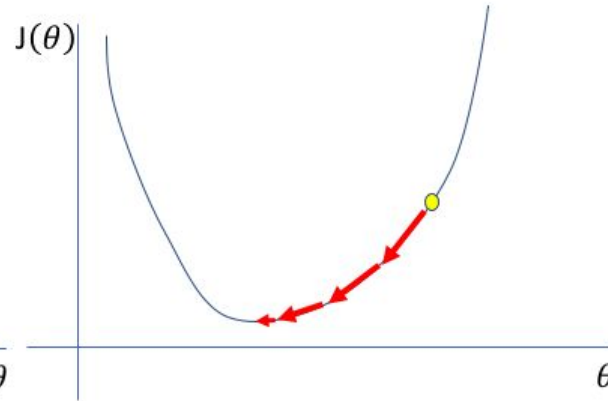
Gradient
i.e. direction of
steepest increase

Too low



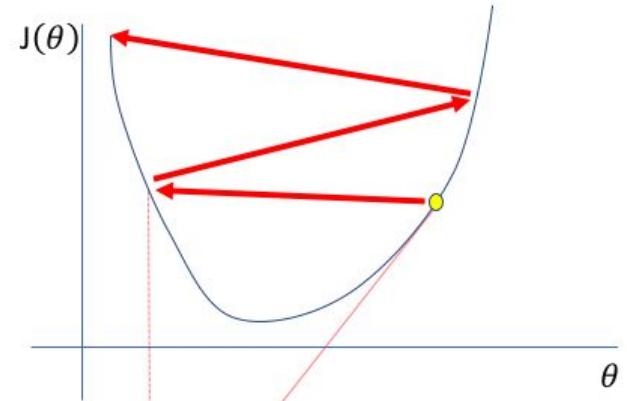
A small learning rate requires many updates before reaching the minimum point

Just right



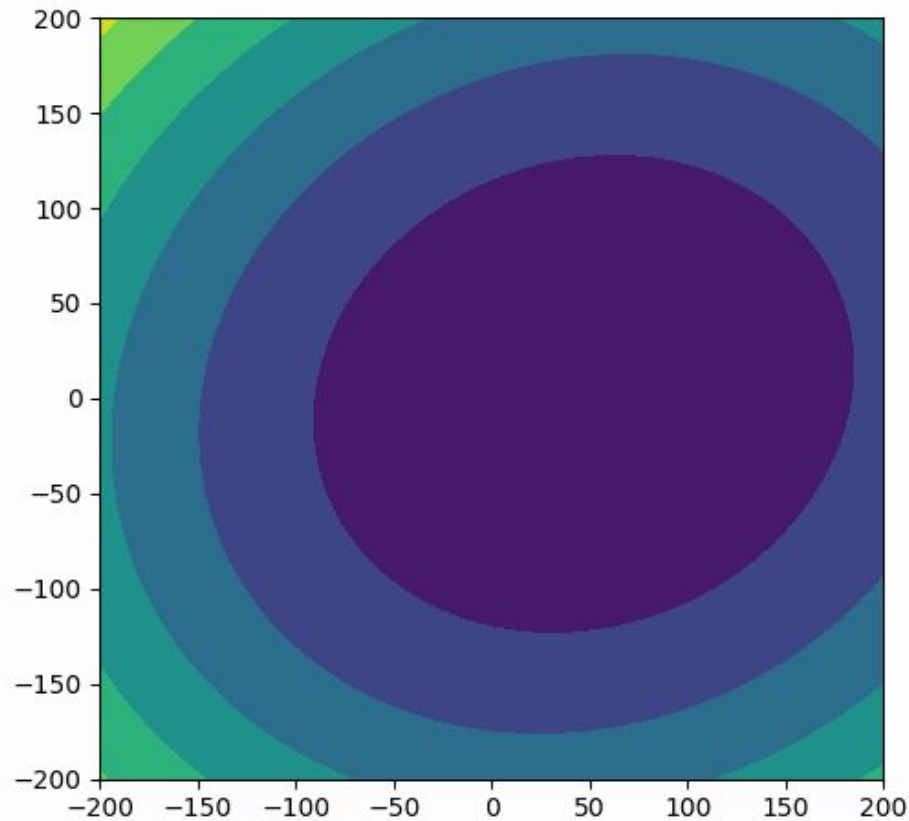
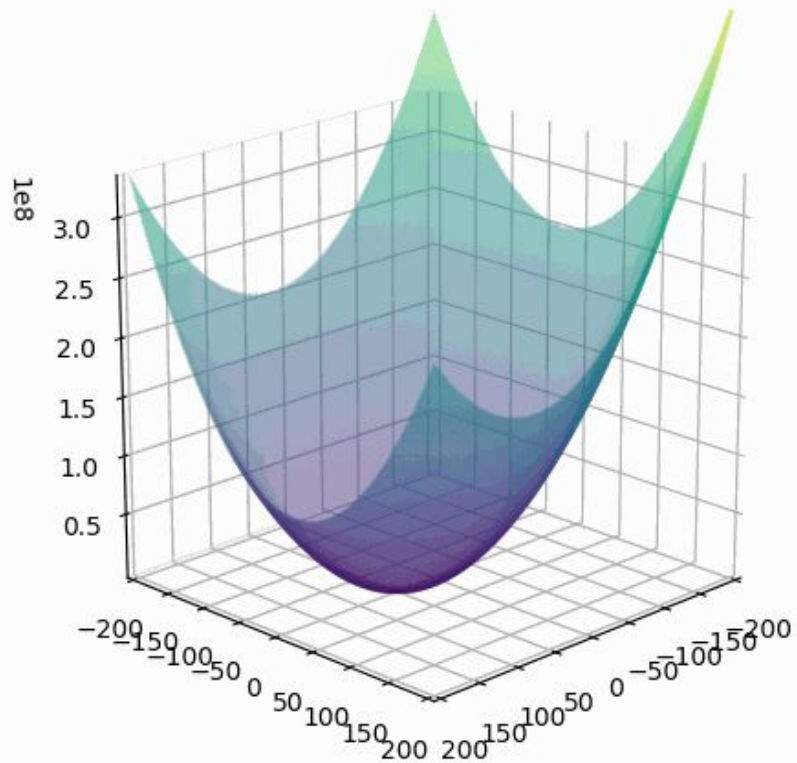
The optimal learning rate swiftly reaches the minimum point

Too high



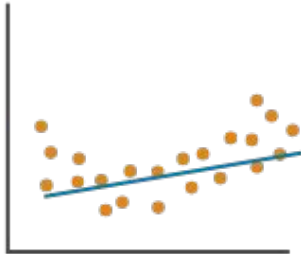
Too large of a learning rate causes drastic updates which lead to divergent behaviors

Ref: <https://www.jeremyjordan.me/nn-learning-rate/>

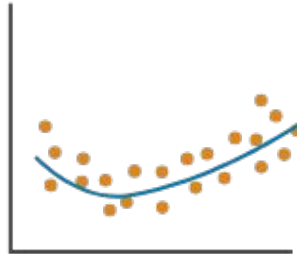


Ref: <https://medium.com/@swayampatil7918/choosing-the-right-regression-loss-function-a-guide-d048b2c317de>

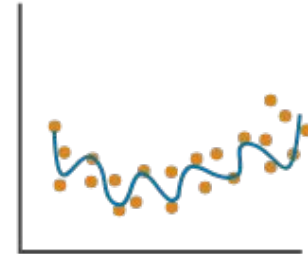
Underfitting



Right fit



Overfitting



Adapted from: <https://www.mathworks.com/discovery/overfitting.html>



Ref: https://www.reddit.com/r/learnmachinelearning/comments/cqb7b8/ml_meme_war_this_is_one_of_my_favourite_machine/

Types of Regression

- Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression
- Decision Tree Regression
- And more...



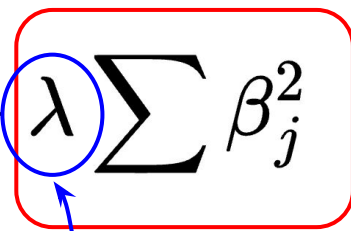
Regularization

Regularization

- Too many features leads to overfitting
- Regularization technique add a penalty to the regression equation
- Ridge and Lasso are two common regularization methods

Ridge Regression

- L2 Regularization
- Adds a penalty on the sum of squared coefficients to the loss function

$$\sum (y_i - \hat{y}_i)^2 + \lambda \sum \beta_j^2$$


- Prevents large coefficients
- Keeps all features but **reduces their impact**
- Works well when many features are correlated

λ controls how strong the penalty is:

- $\lambda = 0 \rightarrow$ normal linear regression
- $\lambda \rightarrow$ weak penalty
- $\lambda \rightarrow$ strong penalty

Large λ forces coefficients to become very small.

Lasso Regression

- L1 Regularization
- Adds a penalty on the sum of absolute values of coefficients

$$\sum (y_i - \hat{y}_i)^2 + \lambda \sum |\beta_j|$$

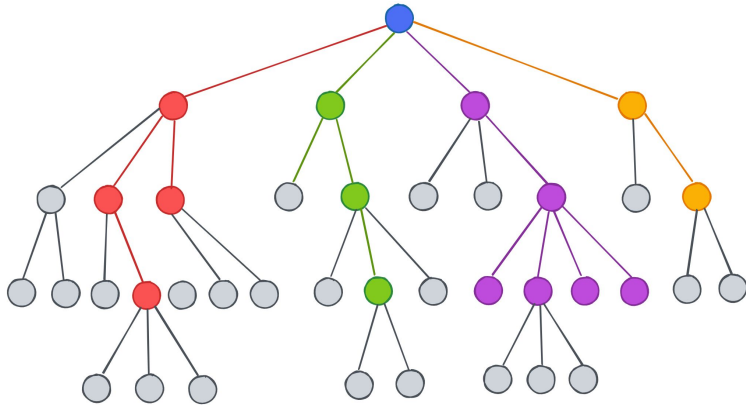
- Feature selection → shrink some coefficients to zero
- Useful when there are irrelevant features in the dataset
- More interpretable than Ridge

Types of Regression

- Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression
- Decision Tree Regression
- And more...

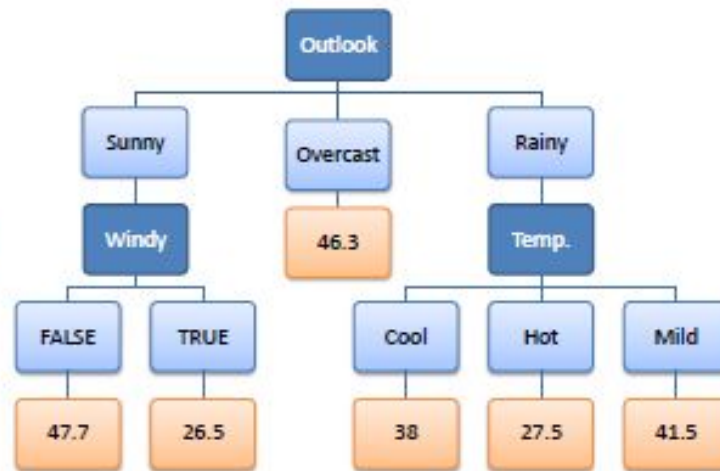
Decision Tree Regression

- ML algorithm for regression tasks
- An internal node represents a decision based on the value of a feature
- A leaf node represents a predicted *value*



- Recursively partitioning the feature space into subsets.
- At each node, the algorithm selects the feature and threshold that result in the greatest reduction in variance of the target variable.
- Measured using metrics such as mean squared error (MSE) or mean absolute error (MAE).

Predictors				Target
Outlook	Temp	Humidity	Windy	Hours Played
Rainy	Hot	High	False	26
Rainy	Hot	High	True	30
Overcast	Hot	High	False	48
Sunny	Mild	High	False	46
Sunny	Cool	Normal	False	62
Sunny	Cool	Normal	True	23
Overcast	Cool	Normal	True	43
Rainy	Mild	High	False	36
Rainy	Cool	Normal	False	38
Sunny	Mild	Normal	False	48
Rainy	Mild	Normal	True	48
Overcast	Mild	High	True	62
Overcast	Hot	Normal	False	44
Sunny	Mild	High	True	30



Ref: https://www.saedsayad.com/decision_tree_reg.htm

Coefficient of Determination R^2

- Measure how well a regression model explains the variation in the data
- The higher the value, the more useful the regression model
- Value ranges between $[0,1]$

$$\begin{aligned} R^2 &= \frac{SS_{yy} - SSE}{SS_{yy}} \\ &= \frac{SS_{yy}}{SS_{yy}} - \frac{SSE}{SS_{yy}} \\ &= 1 - \frac{SSE}{SS_{yy}} \end{aligned}$$

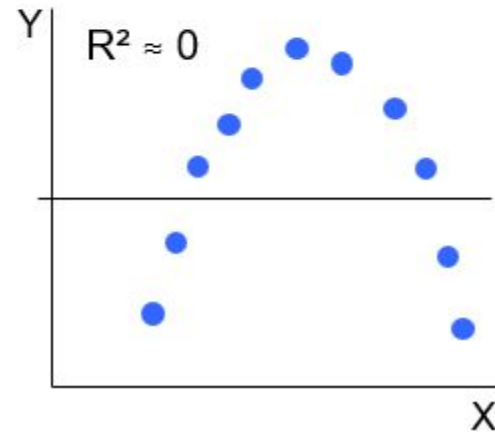
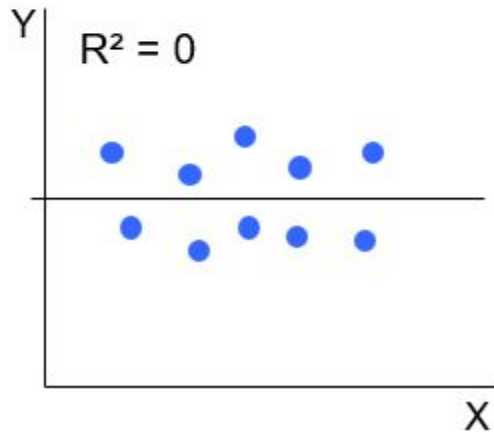
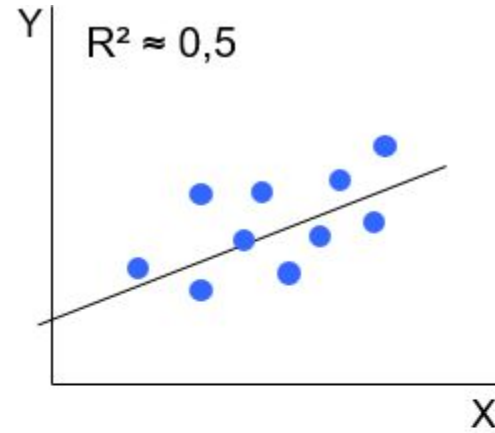
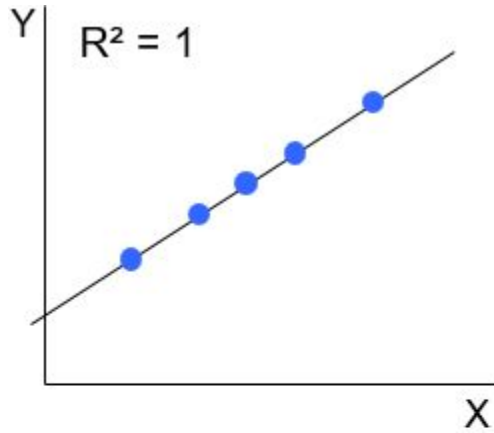
Where:

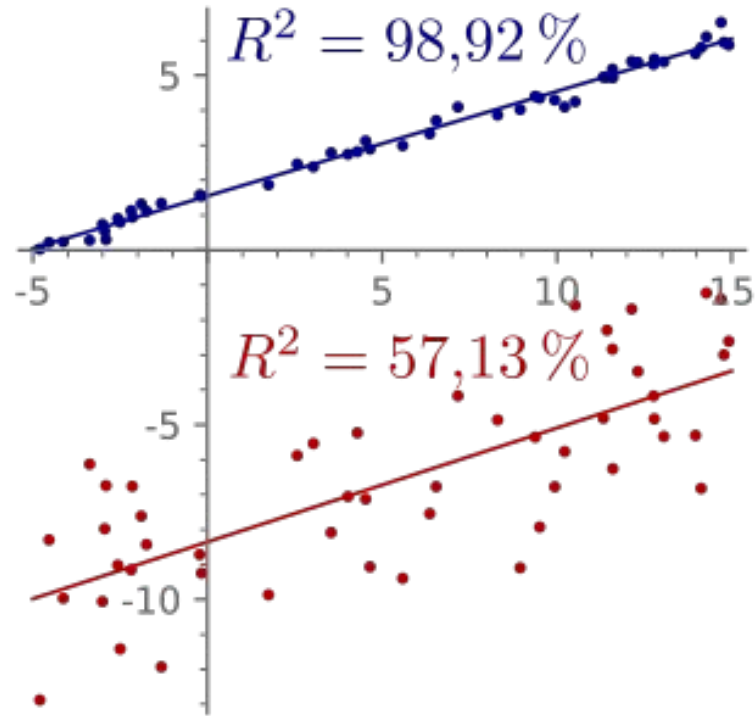
SS_{yy} measures the deviation of the observations from their mean:

$$SS_{yy} = \sum_i (y_i - \bar{y})^2$$

SSE measures the deviation of observations from their predicted values:

$$SSE = \sum_i (y_i - Y_i)^2$$

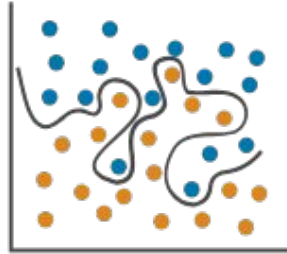




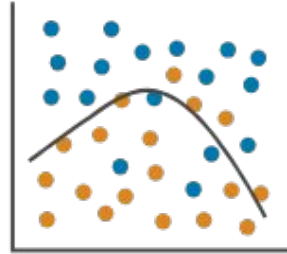
Ref: https://en.wikipedia.org/wiki/Coefficient_of_determination

Classification

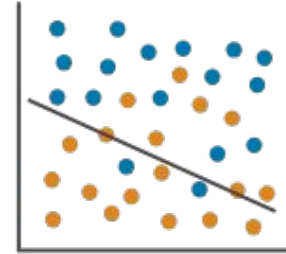
Overfitting



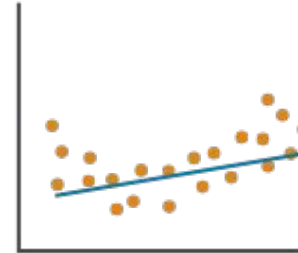
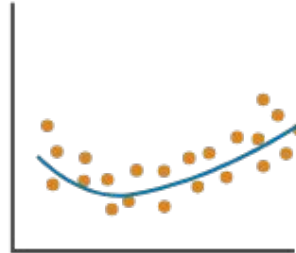
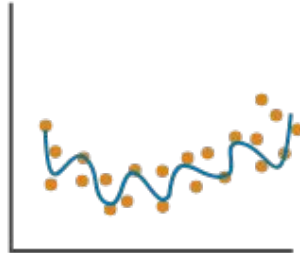
Right Fit



Underfitting



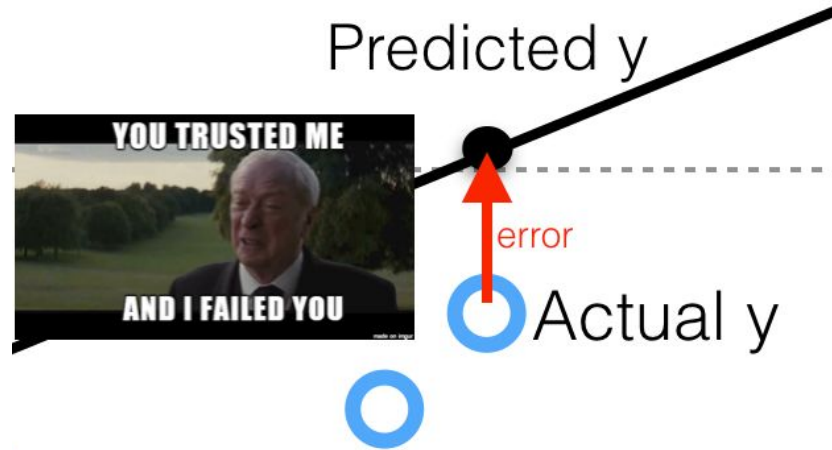
Regression



Ref: <https://www.mathworks.com/discovery/overfitting.html>

In Summary

- Intro to Regression
- Types of Regression
 - Linear
 - Polynomial
 - Ridge
 - Lasso
 - Decision Tree
- Evaluation



Ref: <https://towardsdatascience.com/super-simple-machine-learning-by-me-simple-linear-regression-part-1-concept-and-r-4b5b39bbdb5d>

Q & A